Steering Diffusion Models: EDM Sampling and Classifier-Free Guidance on CIFAR-10 and MNIST

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Abstract

Diffusion models produce great images, but their behavior depends on many tangled design choices. The EDM paper by Karras et al. separates these choices into four clean parts and shows that each one can be studied on its own. In this project we work through a slice of that framework and add classifier-free guidance (CFG), the standard way to steer a diffusion model toward a requested class. We use NVIDIA pretrained EDM models to generate CIFAR-10 images with and without CFG, measure how guidance strength trades off label accuracy and image realism (FID), and train small CFG networks on 100 and 1,000 CIFAR-10 images to see what happens with very little data. We repeat the sweep on MNIST with a small U-Net to confirm the pattern holds across datasets. These experiments set up the Quarter 2 work on continuous diffusion for text generation, where the same EDM design questions reappear in a continuous embedding space.

Code: https://github.com/VedVar43789/EDM_NVIDEA

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1 Introduction

1.1 Problem

Diffusion models are the family behind most of the recent progress in image generation, including Stable Diffusion, Imagen, and DALL-E 2. The idea is simple: start from pure noise and gradually denoise it until a clean image appears. But the details are not. Noise schedule, sampler, preconditioning, and network architecture all interact, and for a long time papers tangled them together. The EDM paper by Karras et al. argued that these choices are mostly independent and can be studied one at a time (Karras et al. 2022). That view is now the standard foundation for sampler design and guidance methods.

This project focuses on one question inside that framework: how do we control what a diffusion model generates, and how do we know the control worked? Our tool is classifier-free guidance (CFG) by Ho and Salimans (Ho and Salimans 2022). CFG mixes two predictions from the denoiser, one with the requested class and one without, and amplifies the difference by a weight w. Larger w pushes harder toward the request but can hurt diversity. The checkpoint goal is to measure that tradeoff concretely.

Beyond Q1, this sets up our Quarter 2 direction. In a meeting this week, Prof. Yusu Wang suggested moving the steering work from images to text generation, specifically using continuous diffusion on token embeddings rather than the usual discrete token approach. The EDM design space carries over directly, since preconditioning, noise schedule, and sampler all come back in a continuous embedding space. The image experiments here are a dress rehearsal for the same questions on text.

1.2 Literature Review and Prior Work

Diffusion models trace back to Sohl-Dickstein et al. (Sohl-Dickstein et al. 2015), who first modeled generation as the reverse of a noising process. Ho et al. formalized the training loss in DDPM (Ho, Jain and Abbeel 2020), and Song et al. unified the field under a stochastic differential equation framework (Song et al. 2021), which also introduced the deterministic probability flow ODE that most fast samplers use today.

Our main reference is EDM (Karras et al. 2022). It separates the design space into four parts: the sampler (step schedule and integrator), the training recipe (noise distribution and loss weighting), the preconditioning (input and output scaling), and the network. The paper shows that smart sampler design can cut the number of network calls per image by 7 to 300 times. On CIFAR-10, EDM reports FID of 1.79 (class-conditional) and 1.97 (unconditional) using only 35 network evaluations per image.

For steering generation, two methods dominate. Classifier guidance by Dhariwal and Nichol (Dhariwal and Nichol 2021) uses the gradient of a separate noisy-image classifier to push samples toward a class, but needs that classifier trained alongside. CFG by Ho and Salimans (Ho and Salimans 2022) avoids this by training one network with the label dropped out some of the time (10 percent in our setup). At sampling time it mixes the conditional and

unconditional predictions with weight w. CFG is now standard in text-to-image diffusion and is the method we study here.

For Quarter 2 on continuous text diffusion, our starting reading list is Diffusion-LM by Li et al. (Li et al. 2022), which embeds tokens into a continuous space and runs Gaussian diffusion there, DiffuSeq by Gong et al. (Gong et al. 2023) for sequence-to-sequence tasks, and Continuous Diffusion for Categorical Data by Dieleman et al. (Dieleman et al. 2022), which adapts the EDM schedule and preconditioning for categorical data.

1.3 Data

CIFAR-10 (Krizhevsky 2009): 60,000 color images at 32x32 across 10 balanced classes (50,000 train, 10,000 test). This is the main dataset for our EDM experiments because NVIDIA released pretrained conditional and unconditional checkpoints for it. We use all 50,000 training images as the FID reference, a pretrained ResNet-56 classifier (94.4 percent accuracy on test) for label accuracy, and class-stratified subsets of 100 and 1,000 training images for our own small training runs.

MNIST (LeCun et al. 1998): 60,000 grayscale handwritten digits at 28x28 across 10 classes. MNIST is our faster testbed, where we train a small class-conditional U-Net from scratch with 10 percent label dropout and evaluate samples with a frozen digit classifier.

Text data for Quarter 2 (planned): We will start on a small, well-studied corpus. Candidates include the E2E restaurant review dataset used in Diffusion-LM (Li et al. 2022), ROCStories, or a small subset of C4 (Raffel et al. 2020). We will decide once we settle on which continuous diffusion method to reproduce.

2 Preliminary Results

All CIFAR-10 experiments ran on a Google Colab A100 GPU. MNIST ran on the same GPU for speed. Every figure and table below is reproducible from the commands in the project README.

2.1 Generation on CIFAR-10

Figure 1 shows 30 unconditional samples from the pretrained EDM model, with recognizable objects across the 10 classes without any class input. Figure 2 shows conditional samples, one class per row, same noise per column. The class picks what is generated and the noise fixes the pose.

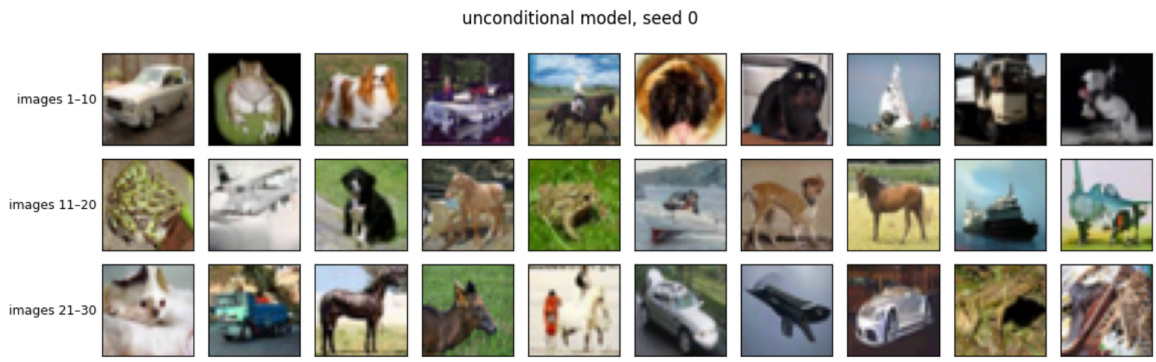


Figure 1: Unconditional CIFAR-10 samples from the pretrained EDM model (seed 0).

2.2 Inside the Sampler and the Role of CFG

Figure 3 shows the EDM sampler at work. At each noise level, the top row is the noisy image and the bottom row is the denoiser's clean-image estimate. At high noise (sigma = 80) the estimate is just a gray blur, and the object becomes recognizable around sigma = 1 to 3. Figure 4 compares the conditional model at $w = 1$ with CFG at $w = 4$, both toward class `cat` and from the same noise. CFG at $w = 4$ locks onto the cat class earlier and produces sharper features at low noise.

2.3 CFG Sweep: Accuracy vs. Realism

Table 1 and Figure 5 are the main checkpoint result. We generate 1,000 images per w (100 per class) with the pretrained EDM models and measure label accuracy, overall FID, and per-class FID (FID_class).

Label accuracy jumps from chance at $w = 0$ to 95 percent at $w = 1$ and saturates near 99 percent after that. Overall FID stays flat until $w = 1.5$ and then climbs, which is expected: CFG shrinks within-class variety so the generated set looks less diverse than the 50,000-image reference. FID_class is lowest at $w = 1.5$ and rises on either side, which gives a visible sweet spot. The two FIDs tell different stories, so we report both.

Our absolute FIDs are much higher than the published EDM numbers (1.79 and 1.97) because EDM uses 50,000 samples and we use 1,000. The two are not directly comparable.

2.4 Training on 100 and 1,000 CIFAR-10 Images

Figure 6 shows two CFG networks trained from scratch on 100 and 1,000 CIFAR-10 images, both for 20,000 updates with 10 percent label dropout. Training loss decreases steadily, but FID of unconditional samples actually gets worse over time (78.7 to 110.8 on the 100-image run, 90.2 to 129.8 on the 1,000-image run).

This is memorization, which the EDM paper predicts in the low-data regime. With so few



Figure 2: Conditional samples, one class per row, same starting noise in each column.

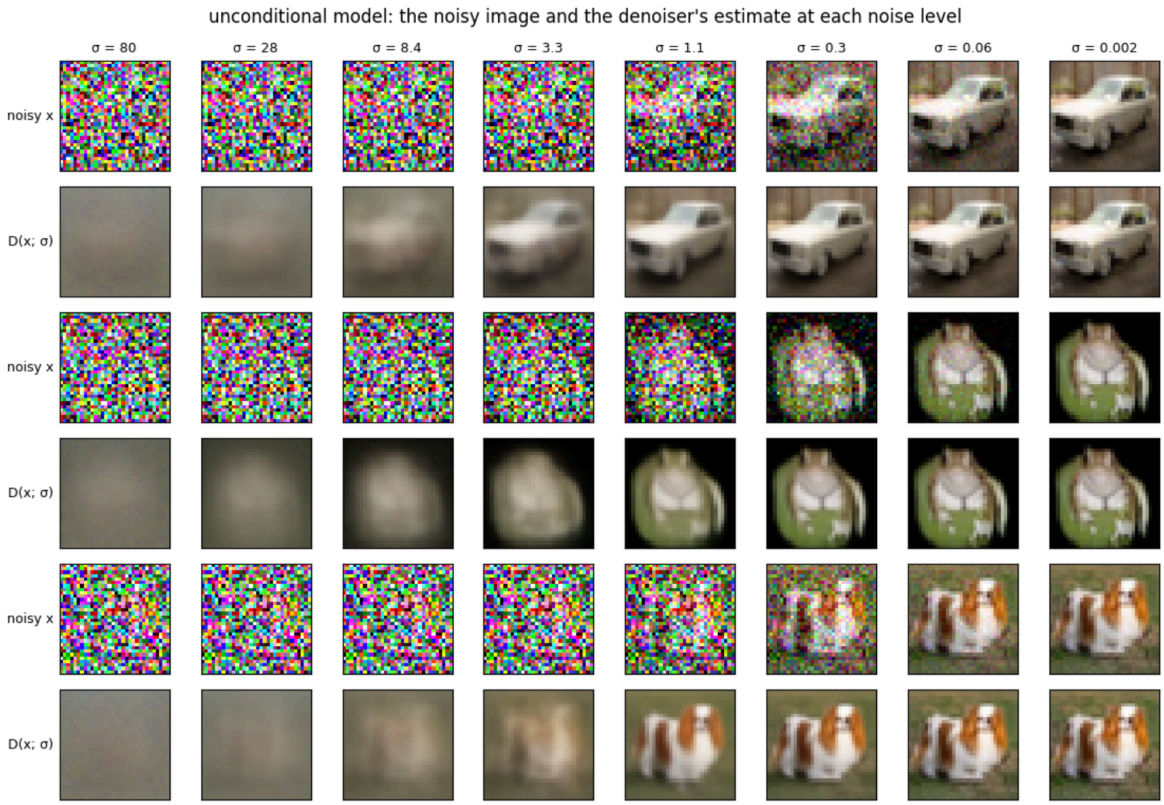
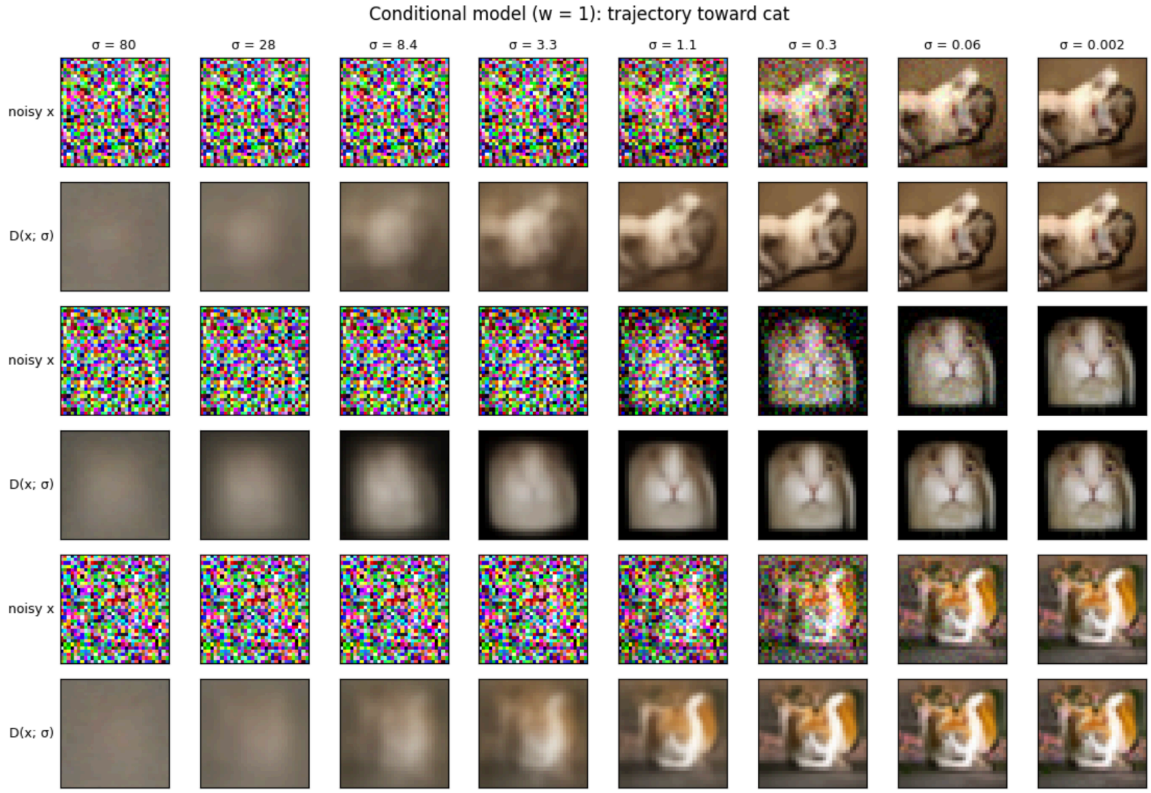


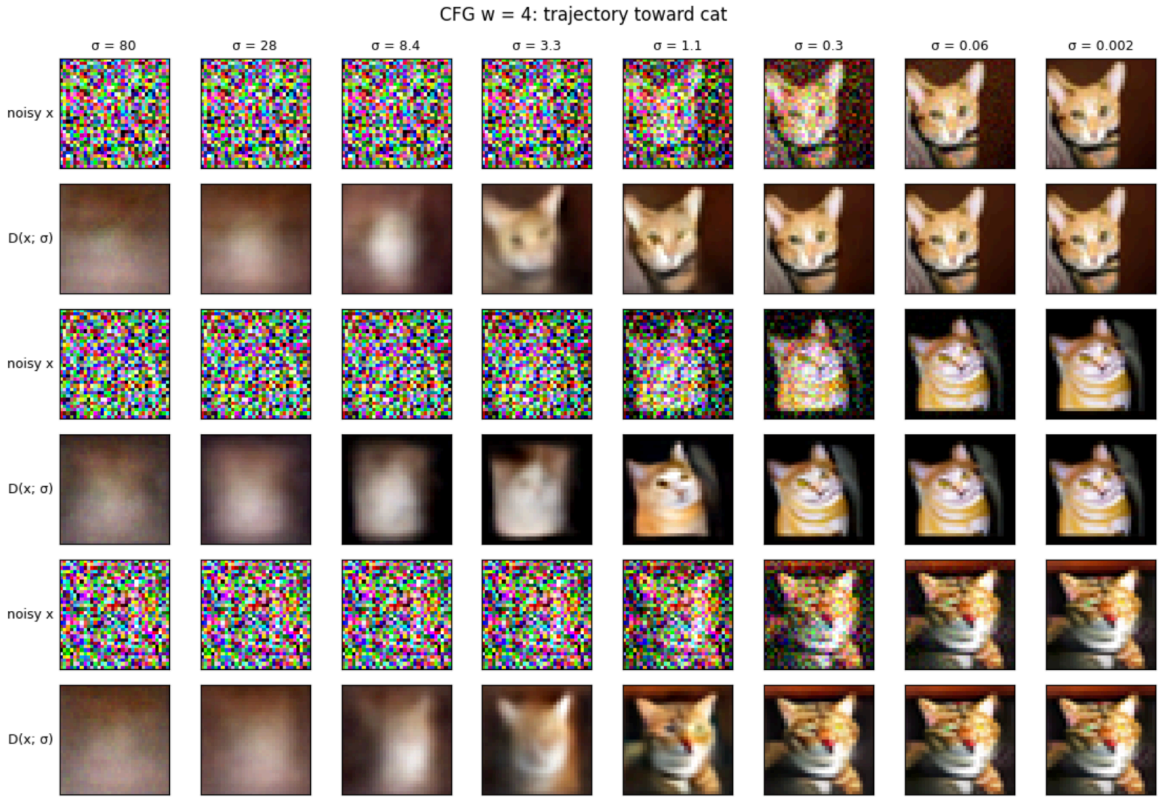
Figure 3: Unconditional sampler trajectory. The denoiser estimate (lower rows) goes from gray blur at high noise to a clear object around sigma = 1 to 3.

Table 1: CFG sweep on CIFAR-10, 1,000 generated images per w, 100 per class.

w	Label Accuracy	FID	FID_class
0	11.5%	31.18	–
1	95.3%	31.33	112.1
1.5	98.7%	31.45	107.6
2	98.8%	33.05	108.9
3	99.5%	36.06	112.5
4	98.9%	39.06	117.6



(a) Conditional model, $w = 1$.



(b) CFG, $w = 4$.

Figure 4: Trajectory toward class cat, same starting noise. Larger w commits to the class earlier.

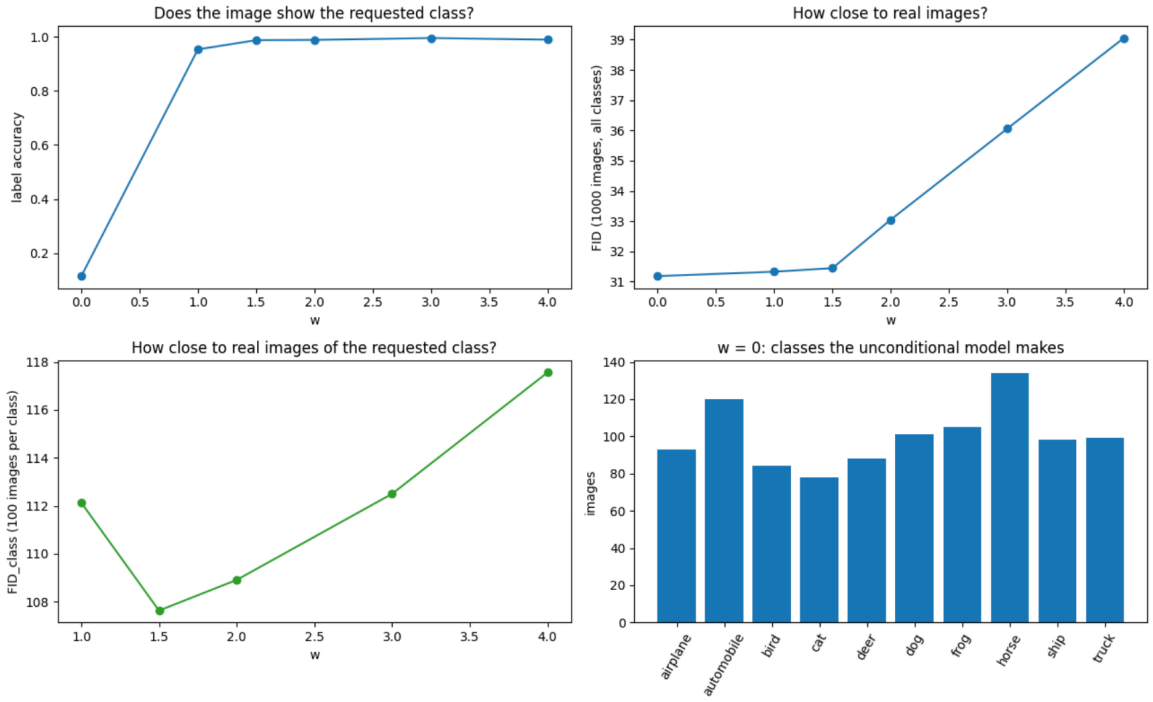


Figure 5: CFG sweep plots. Label accuracy saturates by $w = 1.5$. Overall FID grows past $w = 1.5$. FID_class is lowest at $w = 1.5$.

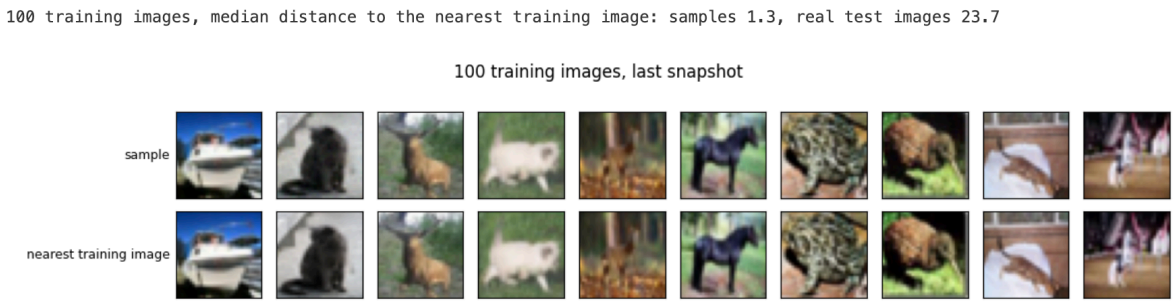
images, the best denoiser essentially copies them. Figure 7 confirms this: samples from the final snapshots match their nearest training images closely. The run is a pipeline test more than a real training run, which is the point of the exercise.



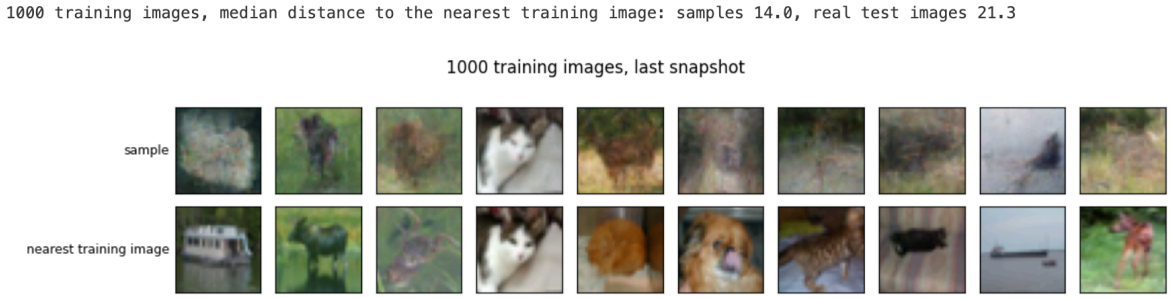
Figure 6: Training loss (left, log scale) and FID across snapshots (right). Loss drops, FID rises: the signature of memorization.

2.5 MNIST CFG Validation

We repeated the sweep on MNIST with a small class-conditional U-Net, 40 samples per w, target digit 3, seed 42. Table 2 and Figure 8 show the result. The pattern matches CIFAR-10: accuracy jumps from chance at $w = 0$ to near-perfect at $w = 1$, and samples at



(a) 100 training images.



(b) 1,000 training images.

Figure 7: Each generated sample (top row of each panel) sits next to its closest training image (bottom row). The match confirms memorization.

$w = 5$ become nearly identical to one another, which is the same loss of diversity we saw on CIFAR-10.

Table 2: MNIST CFG sweep, 40 samples per w, target digit 3, seed 42.

w	Target Match	Confidence	NFE per Image
0	17.5%	0.179	16
1	97.5%	0.986	16
2	100%	1.000	32
5	100%	1.000	32

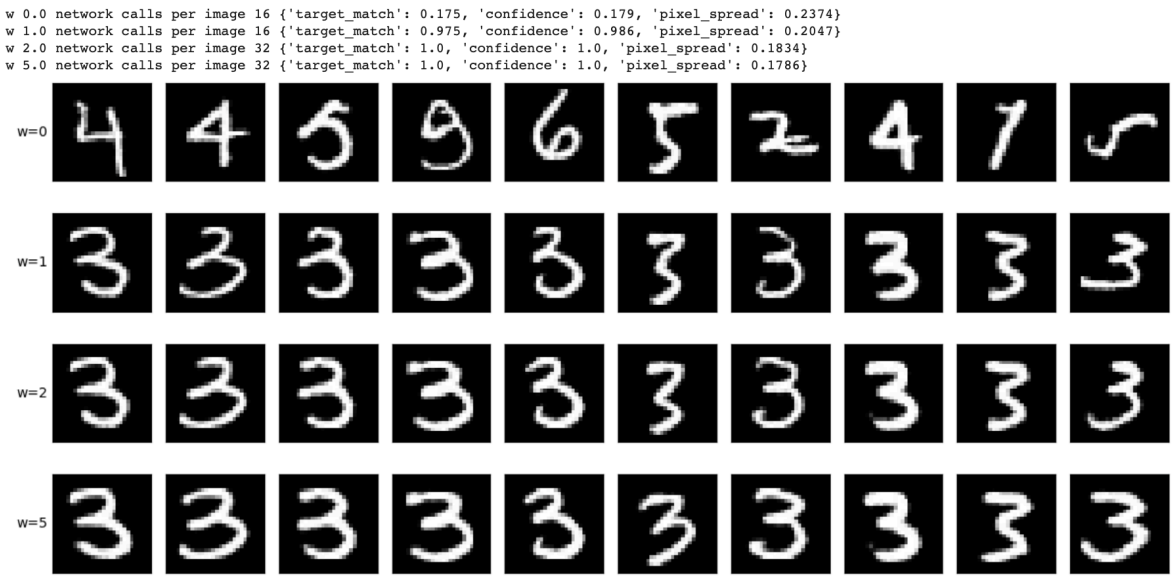


Figure 8: MNIST CFG grid for target digit 3, seed 42. At $w = 0$ the request is ignored. From $w = 1$ onward the samples are all 3s, becoming more uniform as w grows.

2.6 Takeaways

Three things come out of this round of experiments. First, CFG reliably steers samples toward the requested class on both CIFAR-10 and MNIST, with accuracy saturating by $w = 1.5$ or $w = 2$. Second, the accuracy gain costs realism: overall FID grows past $w = 1.5$ and sample diversity visibly drops. Third, small-subset training on CIFAR-10 memorizes rather than generalizes, as the EDM paper predicts.

For Quarter 2 on continuous text diffusion, the same four EDM design axes come back (sampler, training, preconditioning, network), and the accuracy-vs-realism tradeoff here is directly analogous to the fluency-vs-controllability tradeoff that text generation methods have to manage.

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Appendices

A.1 Training Details A1
A.2 Additional Figures A1
A.3 AI Use A1

A.1 Training Details

Expanded for the final submission.

A.2 Additional Figures

Reserved for the final submission.

A.3 AI Use

Per course policy, this section documents how AI assistants (primarily Claude by Anthropic, with Cursor for code edits) were used in this project.

Understanding the assignment. I pasted the course page, Oct 28 TA slides, assignment PDF, and both notebooks into Claude and asked for a consolidated deliverable breakdown. I cross-checked the output against the actual course page to confirm nothing was missed.

Paper summaries. Rather than reading EDM, Diffusion-LM, DiffuSeq, and CDCD cover to cover, I asked Claude for plain-language summaries covering problem, methods, and takeaways. This scoped my Q2 reading list quickly. I verified the EDM summary against the actual paper and corrected anything off.

Code and debugging. The two "Your turn" notebook cells (CIFAR-10 Part 3 trajectory, MNIST Part 3 comparison) were drafted with AI help, then I ran each one and only kept versions that produced sensible figures. I also used AI to explain sampler and training code so I could write about it in my own words.

Writing. My workflow was: write a rough first-pass of each section myself based on my understanding of the experiments and papers, then ask Claude to tighten language and cut redundancy, then rewrite anything that did not sound like me or did not match what I actually did. The content, numbers, section ordering, and main claims are mine. The polishing was AI-assisted. AI also fixed LaTeX issues (empty bib file causing "?" citations, figure path mismatches, subcaption layout).

Organization. I used AI for the initial GitHub scaffolding (folder layout, README skeleton, NOTES.md structure, requirements.txt). The actual notes, predictions, and answers in NOTES.md are my own writing.

Q2 direction. Separate from AI use, I took the initiative this week to set up a meeting with Prof. Yusu Wang and our team to discuss the Q2 project. That meeting led to the pivot from image steering to continuous diffusion for text generation. I used AI afterward to find the three starting papers, but the direction came from the meeting itself.

Verification. Every number in the report is from my own notebook runs, not an AI estimate. Every claim about EDM, CFG, or the Q2 papers was checked against the actual source. Every AI-drafted code cell was executed before being committed. Every citation key in reference.bib maps to a real paper.

Where AI was not used. Running the notebooks, interpreting results, the meeting with Prof. Wang, and maintaining the GitHub repository.